CSE 21 Fa26
Homework 2
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1. (a) How many 7 characte license plates start with a letter or end with a numeral?

Let

$$L = \text{Plates that start with a letter}$$

$$N = \text{plates that end with a numeral}$$

We want to find $|L \cup N|$.

Using the inclusion-exclusion principle,

$$|L \cup N| = |L| + |N| - |L \cap N|$$

There are 26 letter, 10 numerals, and 36 possible character in all.

For plates that start with a letter, the first character has 26 possibilites and the remaining 6 have 36 possibilites.

$$|L| = 26 \cdot 36^6$$

For plates that end with a numeral, the frist 6 character have 36 possibilites and the last has 10 possibilites.

$$|N| = 36^6 \cdot 10$$

For plates that start with a letter and end with a numeral, there are 26 choices for the first character, 10 for the last, and 36 for each of the 5 middle characters.

$$|L \cap N| = 26 \cdot 36^5 \cdot 10$$

Therefore,

$$\boxed{|L \cup N| = 26 \cdot 36^6 + 10 \cdot 36^6 - 26 \cdot 10 \cdot 36^5}$$

(b) How many 7 character license plates use at least one A, at least one B, and at least one numeral?

Let

$$A = \text{Plates without an A}$$

$$B = \text{Plates without a B}$$

$$N = \text{Plates without a numeral}$$

We want a plate containing at least one A, one B, and one numeral.

We use the complement and the inclusion-exclusion principle. There are 36^7 possible plates. The individual sets have sizes

$$|A| = 35^7, \qquad |B| = 35^7$$

$$|N| = 26^7$$

Their pairwise intersections are

$$|A \cap B| = 34^7$$

$$|A \cap N| = 25^7$$

$$|B \cap N| = 25^7$$

The interesection of all three sets is

$$|A \cap B \cap N| = 24^7$$

Using inclusion-exclsuion

$$\begin{aligned} |A \cup B \cup N| = |A| + |B| + |N| \\ - |A \cap B| \\ - |A \cap N| \\ - |B \cap N| \\ + |A \cap B \cap N| \end{aligned}$$

We subtract this from the total possibilites:

$$\boxed{36^7 - \left(2(35^7) + 26^7 - 34^7 - 2(25^7) + 24^7\right)}$$

(c) How many 7 character license plates contain only numerals that add up to 10?
We need to the find the number of solutions to

$$x_1 + x_2 + \cdots + x_7 = 10$$

where

$$0 \leq x_i \leq 9$$

We can use the stars and bars formula to count the solutions

$$\binom{n+k-1}{k-1}$$

Here, $n = 10$ and $k = 7$.

$$\binom{10+7-1}{7-1} = \binom{16}{6}$$

However, stars and bars allows individual values greater than 9.
Since the toal sum is 10, the only invalid cases are there where one digit equals 10 and the remaining 6 equal 0.
For example,

$$(10, 0, 0, 0, 0, 0, 0)$$

$$(0, 10, 0, 0, 0, 0, 0)$$

$$\vdots$$

There are exactly 7 invalid cases, one for each possible position of 10. We subtract these cases:

$$\boxed{\binom{16}{6} - 7}$$

(d) How many 7 character license plates contain only letters in alphabetical order, with repitions allowed?

We use combination with repition:

$$\binom{n+k-1}{k}$$

where

$$n = 26 \text{ letters}$$

$$k = 7 \text{ characters}$$

Because the letters must appear in alphabetical order, each selection of 7 letters corresponds to exactly one valid license plate.

Substituting

$$\binom{26+7-1}{7} = \binom{32}{7}$$

Therefore,

$$\boxed{\binom{32}{7}}$$

2. (a) How many different ways cam the museum curator arrange all 9 prints among the 12 rooms?

We have 9 identical prints and 12 different rooms.

We use stars and bars because we are distrbuting identical objects among distinct groups.

Let x_i represent the number of prints placed in room i.

We need

$$x_1 + x_2 + \cdots + x_{12} = 9$$

where $x_i \geq 0$.

Using stars and bars,

$$\binom{n+k-1}{k-1}$$

where $n = 9$ and $k = 12$.

$$\binom{9+12-1}{12-1} = \binom{20}{11}$$

Therefore,

$$\boxed{\binom{20}{11}}$$

(b) How many ways can the museum curator arrange all 9 prints among the 12 rooms so that each print is in a different room?

Since all 9 prints are identical and must be placed in different rooms, we only need to choose which 9 of the 12 rooms recieve a print, regardless of order.

Therefore,

$$\boxed{\binom{12}{9}}$$

(c) How many ways can the museum curator arrange all 9 prints among the 12 rooms so that at least 3 rooms are non-empty?

We can use complementary countin.

We begin with all possibile distributions and subtract the ones where only 1 or 2 rooms are non-empty.

From part (a), the total number of distributions is:

$$\binom{20}{11}$$

First, consider exactly 1 non-empty room.

We chhose 1 of the 12 rooms to contain all 9 prints

$$\binom{12}{1} = 12$$

Next consider exactly 2 non-empty rooms.

We first chhose 2 rooms from the 12

$$\binom{12}{2}$$

We then distribute the 9 identical prints between those two

Let x_1 and x_2 represent the number of prints in each room.

$$x_1 + x_2 = 9$$

where $x_1, x_2 \geq 1$.

using stars and bars:

$$\binom{9-1}{2-1} = \binom{8}{1}$$

Thus, the number of distribution with exactly 2 nonempty rooms is

$$\binom{12}{2}\binom{8}{1}$$

Subtracting both invalid cases from the total gives

$$\boxed{\binom{20}{11} - 12 - \binom{12}{2}\binom{8}{1}}$$

(d) How many ways can the museum curator arrange all 9 prints among the 12 rooms such that at least one room has at least 3 prints?

We can use complementary counting.

We begin with all possible distributions and subtract the ones where every room has at most 2 prints.

From part (a), the total number of distributions is:

$$\binom{20}{11}$$

First, consider one room with at least 3 prints.

We choose 1 of the 12 rooms and put 3 prints in it. The other 6 prints can go in any of the 12 rooms.

using stars and bars:

$$\binom{6+12-1}{12-1} = \binom{17}{11}$$

So the number of distributions with one fixed room getting at least 3 prints is

$$\binom{12}{1}\binom{17}{11}$$

Next consider two rooms with at least 3 prints each.

We choose 2 of the 12 rooms and put 3 prints in each. That uses 6 prints, so 3 are left for the 12 rooms.

using stars and bars:

$$\binom{3+12-1}{12-1} = \binom{14}{11}$$

So this case is

$$\binom{12}{2}\binom{14}{11}$$

Next consider three rooms with at least 3 prints each.

We choose 3 of the 12 rooms and put 3 prints in each. That uses all 9 prints, so there is only 1 way to place what is left.

$$\binom{12}{3}$$

We can't have 4 rooms with at least 3 prints each, since that would take 12 prints.

Using inclusion-exclusion, the number of distributions where every room has at most 2 prints is

$$\binom{20}{11} - \binom{12}{1}\binom{17}{11} + \binom{12}{2}\binom{14}{11} - \binom{12}{3}$$

Subtracting this from the total gives

$$\boxed{\binom{12}{1}\binom{17}{11} - \binom{12}{2}\binom{14}{11} + \binom{12}{3}}$$

(e) How many ways can the museum curator arrange all 9 prints in 3 of the 12 rooms such that each room has exactly 3 prints?

We need to choose exactly 3 rooms from the available 12.

Since the 9 prints are identical, once we select the 3 rooms, there is only one way to place exactly 3 prints in each room.

Therefore, we use combination to choose the 3 rooms:

$$\boxed{\binom{12}{3}}$$

3. Compute the number of integer solutions to

$$a_1 + a_2 + a_3 + a_4 + a_5 + a_6 = 30.$$

(a)

$$1 \le a_1, \quad 2 \le a_2, \quad 3 \le a_3, \quad 4 \le a_4, \quad 5 \le a_5, \quad 6 \le a_6.$$

First, assign each variable its minimu required value.

$$1 + 2 + 3 + 4 + 5 + 6 = 21.$$

This leave $30 - 21 = 9$ remaining. Let $x_i = a_i - i$ for $1 \le i \le 6$. Then

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 9,$$

where each $x_i \ge 0$.

Using 9 stars and 5 bars the number of solutions is:

$$\boxed{\binom{9+6-1}{6-1} = \binom{14}{5}}.$$

(b)

$$a_1 + a_2 + a_3 + a_4 + a_5 + a_6 = 30,$$

where at least one variable is less than or equal to 3.

We use the complement method by subtracting the solution where every variable is at least 4 from the total number of solution

Using Stars and Bars, the total number of solution without any restriction is:

$$\binom{30+6-1}{6-1} = \binom{35}{5}.$$

For the complement, each variable must satisfy $a_i \ge 4$. Assigning 4 to each variable uses:

$$6 \cdot 4 = 24,$$

leaving $30 - 24 = 6$.

Let $x_i = a_i - 4$. Then

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 6,$$

where each $x_i \ge 0$

By Stars and Bars, the complement solutins is:

$$\binom{6+6-1}{6-1} = \binom{11}{5}.$$

Subtracting these from the total gives the valid solutions:

$$\boxed{\binom{35}{5} - \binom{11}{5}}.$$

(c)

$$a_1 + a_2 + a_3 + a_4 + a_5 + a_6 = 30,$$

where

$$0 \leq a_i \leq 7 \quad \text{for all } 1 \leq i \leq 6.$$

Use Inclusion-Exclusion to subtract the solutions where at least one variable is greater than or equal to 8.

No restrictions, using Stars and Bars:

$$\binom{35}{5}$$

solutions.

Let A_i be the set of solutions where $a_i \geq 8$.

If one variable is at least 8, we assign 8 to that variable, leaving $30 - 8 = 22$. There are $\binom{6}{1}$ ways to select the variable:

$$\binom{6}{1}\binom{22+5}{5} = \binom{6}{1}\binom{27}{5}$$

Now if two variables are at least 8 we are left with $30 - 16 = 14$. There are $\binom{6}{2}$ ways:

$$\binom{6}{2}\binom{14+5}{5} = \binom{6}{2}\binom{19}{5}.$$

Moving on to if three variable are at least 8 we are left with $30 - 24 = 6$. There are $\binom{6}{3}$ ways:

$$\binom{6}{3}\binom{6+5}{5} = \binom{6}{3}\binom{11}{5}.$$

This is the max case because if four variables are at least 8 we get $4 \cdot 8 = 32 > 30$.

Therefore the number of valid solutions is given by:

$$\boxed{\binom{35}{5} - \binom{6}{1}\binom{27}{5} + \binom{6}{2}\binom{19}{5} - \binom{6}{3}\binom{11}{5}}.$$

4. (a) For $n \geq 0$ and $k \geq 0$, consider the identity

$$k\binom{n}{k} = n\binom{n-1}{k-1}.$$

Prove it combinatorially by counting the same set in two different ways.

Imagine we have n basketball players. We want to form a team containing k players, and we must select one to be the team captain.

LHS: First choose k players from the n players. There are $\binom{n}{k}$ ways. Then choose a team captain from the k selected players. There are k possible captains.

$$\text{LHS} = \binom{n}{k} \cdot k.$$

RHS: First choose the captain from all n players. There are n ways to do this. Now choose the remaining team members. Since a captain has already been chosen, there are $n-1$ players left, and we still need $k-1$ more players (co-captain, point guard, and so on). There are $\binom{n-1}{k-1}$ ways to do this.

$$\text{RHS} = n\binom{n-1}{k-1}.$$

Both sides count the same teams with a captain, so

$$k\binom{n}{k} = n\binom{n-1}{k-1}.$$

(b) For $n \geq 3$, prove that

$$\binom{n+1}{3} = 1(n-1) + 2(n-2) + 3(n-3) + \cdots + (n-3)3 + (n-2)2 + (n-1)1$$

using a combinatorial argument.

[Hint: count the fixed density binary strings with 3 ones based on where the middle one is.]

We look at binary strings with $n+1$ bits and exactly three 1s. Choose 3 bits to be ones. The rest are zeros:

$$n + 1 - 3 = n - 2$$

zeros. So there are $\binom{n+1}{3}$ such strings.

Call the earliest 1 B, the middle 1 M, and the latest 1 E. B is somewhere in positions 1 through $M-1$, E is somewhere in positions $M+1$ through $n+1$, and M itself has to be in one of the positions $2, 3, \ldots, n$.

For a fixed position of M,

$$B: \ \binom{M-1}{1} = M - 1$$

ways to choose its position, and

$$E: \ \binom{n+1-M}{1} = n + 1 - M$$

ways to choose its position. So that middle position gives

$$(M-1)(n+1-M)$$

strings.

Now let M run from 2 to n:

$$\begin{aligned}\binom{n+1}{3} &= (2-1)(n+1-2) + (3-1)(n+1-3) + (4-1)(n+1-4) + \cdots + (n-1)(n+1-n) \\ &= 1(n-1) + 2(n-2) + 3(n-3) + \cdots + (n-1)1.\end{aligned}$$